

0.1 Section 7.2 - Probability Theory

We didn't actually go over much

0.1.1 Event Independence

Definition 1 (Event Independence)

If H and Q are events, we say H and Q are **independent** if and only if

$$P(H \cap Q) = P(H) \cdot P(Q)$$

Example 1

A card is selected from a standard 52 deck. Determine the probability that

1. The card is a heart

$$\begin{aligned} P(\heartsuit) &= \frac{\# \text{ hearts}}{\# \text{ cards}} \\ &= \frac{13}{52} \\ &= \boxed{\frac{1}{4}} \end{aligned}$$

2. The card is a queen

$$\begin{aligned} P(\text{Queen}) &= \frac{\# \text{ queens}}{\# \text{ cards}} \\ &= \frac{4}{52} \\ &= \boxed{\frac{1}{13}} \end{aligned}$$

3. The card is a heart and a queen

$$\begin{aligned} P(\heartsuit \text{ and queen}) &= \frac{\# \text{ heart and queen}}{\# \text{ cards}} \\ &= \frac{1}{52} \end{aligned}$$

Notice that

$$P(\heartsuit \text{ and queen}) = P(\heartsuit) \cdot P(\text{queen})$$

This means that H and Q are **independent**

I would put the examples leading up to the derivation of this next thing but I'm too lazy

0.1.2 Binomial Probability

Definition 2 (Binomial Probability)

Given a limited event (ie 2 outcomes) with probability p , Then

$$P(k \text{ successes in } n \text{ trials}) = \binom{n}{k} p^k \cdot (1-p)^{n-k}$$

Example 2

Let's do some examples

A biased coin has probability of 0.51 for landing heads when flipped. Determine the probability of the coin landing on heads exactly 6 times when the coin is flipped 10 times

$$\begin{aligned} p &= 0.51 \\ k &= 6 && \text{(successes)} \\ n &= 10 \end{aligned}$$

$$\begin{aligned} P(6 \text{ successes in } 10 \text{ trials}) &= \binom{10}{6} (0.51)^6 \cdot (1 - 0.51)^{10-6} \\ &\approx \boxed{0.213} \end{aligned}$$

Example 3

Let's do another

Determine the probability of rolling a 12 **at least once** when a pair of dice is rolled 20 times.

Since we have "at least" let's do the opposite

$$P(\text{never rolling 12 out of 20}) = \binom{20}{0} \left(\frac{1}{36}\right)^0 \cdot \left(\frac{35}{36}\right)^{20} \\ \approx 0.569\dots$$

$$P(\text{rolling 12 at least once out of 20}) = 1 - P(\text{never rolling 12 out of 20}) \\ \approx 1 - 0.569 \\ \approx \boxed{0.431}$$